

Sine, cosine, and tangent ratios



1 $\angle ACB$

2 a $\frac{24}{7}$

b $\frac{12}{5}$

3 $\frac{4}{5}$

4 a $\frac{5}{13}$

b $\frac{12}{13}$

5 a $\frac{7}{25}$

b $\frac{7}{25}$

6 a $\frac{4}{5}$

b $\frac{4}{5}$

7 a 21 : 23
c 23 : 31.14

b 21 : 23
d 63 : 93.43

8 $\frac{\text{opposite}}{\text{hypotenuse}}$

9 a $\sin \theta = \frac{AC}{AB}$
c $\tan \theta = \frac{AC}{BC}$

b $\cos \theta = \frac{BC}{AB}$

10 a $\tan \theta = \frac{8}{15}$

b $\cos \theta = \frac{15}{17}$

11 a $\tan \theta = \frac{3}{4}$

b $\sin \theta = \frac{4}{5}$



12

a $\cos \theta = \frac{5}{13}$

c $\tan \theta = \frac{7}{4}$

b $\sin \theta = \frac{24}{25}$

d $\tan \theta = \frac{24}{7}$

15

a $\angle BCA$

c $\tan \theta = \frac{4}{3}$

b $\cos \theta = \frac{3}{5}$

16

a $\angle BCA$

c $\sin \theta = \frac{4}{5}$

b $x = 8$

d $\tan \theta = \frac{4}{3}$

17

a $x = 13$

c $\cos \theta = \frac{5}{13}$

b $\sin \theta = \frac{12}{13}$

18

a i $x = 4$

ii $\tan \theta = \frac{4}{3}$

b i $x = 12$

ii $\tan \theta = \frac{5}{12}$

19

a Yes

c No

b Yes

20 $\frac{3.5}{25} \leq \tan x \leq \frac{5}{25}$



Additional questions

21

a i BC ii AC iii BA

c i Y ii Z iii X

b

i X ii Y iii Z

d

i b ii a iii c

22

a $\angle C$

c $\angle B$

b $\angle B$

d $\angle A$

23

a AC

c BC

e $\angle ACB$

b BC

d AC

24

a i $\frac{21}{29}$ ii $\frac{20}{29}$ iii $\frac{21}{20}$

c i $\frac{5}{7}$ ii $\frac{4.9}{7}$ iii $\frac{5}{4.9}$

e i $\frac{9}{31}$ ii $\frac{29.7}{31}$ iii $\frac{9}{29.7}$

b i $\frac{12}{13}$ ii $\frac{5}{13}$ iii $\frac{12}{5}$

d i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$

f i $\frac{21}{29}$ ii $\frac{20}{29}$ iii $\frac{21}{20}$

25

a $\angle BCA$

c $\frac{15}{17}$

b $\frac{8}{17}$

26 From the diagram we have the following relationships: $\sin \theta = \frac{12}{13}$, $\cos \theta = \frac{5}{13}$ $\tan \theta = \frac{12}{5}$.

$$\frac{\sin \theta}{\cos \theta} = \frac{\frac{12}{13}}{\frac{5}{13}} = \frac{12}{13} \times \frac{13}{5} = \frac{12}{5} \text{ which is equal to } \tan \theta \text{ as required.}$$

27 a $\frac{BC}{AB}$
 c $\frac{BC}{AC}$



$\frac{AC}{AB}$

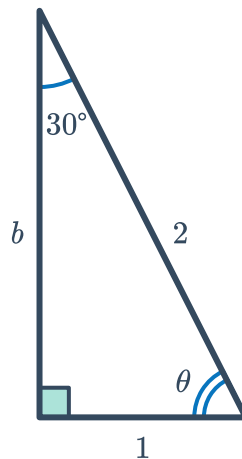
28 We can see that $\sin \theta = \frac{3}{5}$ and $\cos \alpha = \frac{3}{5}$.

This means $\sin \theta = \cos \alpha$. This is because the opposite side in reference to θ is the adjacent side in reference to α and the hypotenuse must be the same.

29 The two triangles are similar triangles as enlargement is a similarity transformation. This means that their side lengths are all in the same ratio.

As an example, let's look at the right triangle with side lengths 3, 4, 5. If $\cos \theta = \frac{3}{5}$ then the cosine ratio of the corresponding angle would be equal to $\frac{12}{20} = \frac{3}{5}$ which is the same value as $\cos \theta$ in the smaller triangle. We can prove the sine and tangent ratios are equal in a similar manner.

30



Any right triangle that has an angle measure of 30° must have an angle with a measure of 60° . This means it is a 30° - 60° - 90° triangle, and the ratio of sides in a 30° - 60° - 90° triangle is $1 : \sqrt{3} : 2$. The side opposite the angle with measure of 30 must be the shortest side of the triangle, and the adjacent side must be the middle length. This means $\tan 30^\circ = \frac{1}{\sqrt{3}}$, so by

rationalizing the denominator $\tan 30^\circ = \frac{\sqrt{3}}{3}$